

GLV Management System

Detailed System Documentation and Operator Reference

Document	System documentation
Prepared for	God's Love Ventures / GLV Management System operators
Generated	September 09, 2026
Workspace	C:\Users\andre\glv-management-system
Stack	Next.js 16 App Router, React 19, Auth.js v5, Prisma, Neon Postgres, Tailwind CSS

This document describes the current GLV Management System as implemented in the repository. It intentionally avoids secrets and shows environment variable names only.

1. Executive Overview

GLV Management System is a business operations application for managing layaway customers, product accounts, installment payments, staff, procurement readiness, credits/refunds, reports, settings, notifications, audit logs, and admin support. The system is built as a server-rendered Next.js App Router application with Prisma-backed persistence in Neon Postgres.

Area	What it does
Customer operations	Registers customers, assigns staff, creates product accounts, records payments, tracks balances, and keeps customer/account histories together.
Financial operations	Records payments, generates receipt numbers, recalculates balances, creates overpayment credits, tracks salaries, and exports weekly Excel reports.
Procurement operations	Shows products that should be bought when linked accounts reach at least 70% paid, including fully paid but undelivered accounts.
Staff operations	Manages staff records, applications, salaries, permissions, login accounts, password reset, and staff-specific dashboards.
Control and governance	Uses role/permission gates, audit logs, admin confirmations, settings, lifecycle rules, and protected server actions.

2. High-Level Architecture

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Layer	Key files	Responsibility
Presentation shell	app/layout.tsx, components/app-shell.tsx, components/dashboard-nav.tsx	Protected layout, sidebar navigation, role-specific menu visibility, notification badges, admin AI support bubble.
Pages	app/**/page.tsx	Server-rendered module screens for dashboard, customers, accounts, payments, products, staff, reports, settings, activity, credits, and audit logs.
Mutations	actions/*.ts	Server Actions for authenticated business writes and audit logging.
Domain services	lib/*.ts	Shared business logic for auth config, reports, lifecycle, procurement, account creation, payment recording, settings, permissions, and utilities.
Persistence	prisma/schema.prisma, prisma/migrations	Data model and migration history for Neon Postgres.
API routes	app/api/**/route.ts	Auth route, change-password, logout, notifications, weekly export, system health, and AI support.

3. Data Model and Relationships

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Model	Purpose	Important fields
User	Login identity and role/permission carrier.	email, password, role, staffId, permissions, mustChangePassword, online, lastSeenAt
Staff	Employee profile connected to customers, salaries, and optional user.	code, fullName, email, phone, active, monthlySalary
StaffApplication	Public staff signup request awaiting admin review.	fullName, email, phone, status, reviewedBy, reviewedAt
Customer	Person buying products through layaway.	customerId, fullName, phone, address, nationalId, staffId
Product	Sellable product or combo.	name, category, costPrice, transportCost, layawayPrice, dailyAmount, duration, quantityOnSale, active
CustomerAccount	A customer's product layaway contract.	targetAmount, dailyAmount, totalPaid, balance, status, deliveryStatus, deliveredAt
Payment	Installment payment record with generated receipt.	receiptNo, accountId, amount, paymentDate, method, receivedBy
CustomerCredit	Overpayment or closure refund credit.	amount, remainingAmount, status, source, accountId, paymentId
StaffSalaryPayment	Payroll payment tracking.	staffId, amount, paymentDate, notes, paidBy
AuditLog	Immutable-style activity history for key changes.	userId, action, entity, entityId, oldValue, newValue
Setting	Company, layaway, payroll, security, notification, theme, and system configuration.	companyName, thresholds, prefixes, payrollDay, permissions-related defaults, status fields

4. Roles, Permissions, and Authentication

Authentication uses Auth.js v5 credentials flow. Route authorization is centralized in the Auth.js authorized callback. Users with mustChangePassword are redirected to /change-password until they complete the first-login reset. Staff are limited to operational routes unless granted explicit permissions.

Role / permission	Access pattern
SUPER_ADMIN	Full administrative role. Treated as admin throughout the UI and server checks.
ADMIN	Administrative access, including settings and privileged modules.
STAFF	Operational access to dashboard, activity, assigned customers/accounts/payments, and change-password.

MANAGE_CUSTOMERS	Allows broader customer management beyond assigned staff ownership.
MANAGE_ACCOUNTS	Allows account creation/correction/delivery operations beyond assigned ownership.
MANAGE_PAYMENTS	Allows payment management, credits/refunds, and privileged payment workflows.
VIEW_REPORTS	Allows access to reports and weekly export.
MANAGE_PRODUCTS	Allows product catalog and procurement module access.
MANAGE_STAFF	Allows staff records, applications, salaries, and password resets.
VIEW_AUDIT_LOGS	Allows read-only audit log access.

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stores id, role, permissions, staffId, mustChangePassword</w:t></w:r></w:p><w:p><w:pPr><w:pStyle
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route</w:t></w:r></w:p>
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5. Module Catalog

Module	Route(s)	What it does
Public Home	/	Landing/public entry point for GLV.
Login	/login	Credentials login with Auth.js server action integration.
Forgot Password	/forgot-password	Password reset request UI.
Signup	/signup	Staff application submission before admin approval.
Change Password	/change-password,	First-login or forced password change; clears stale auth cookies

	/api/change-password	after success.
Dashboard	/dashboard	Role-shaped summary: admins see business KPIs; staff see assigned operational metrics.
Activity	/activity	Operational activity and collection trends.
Customers	/customers, /customers/new, /customers/[id], /customers/[id]/edit	Customer CRUD, search/filtering, assignment, profile, account/payment history, and customer-level actions.
Accounts	/accounts, /accounts/new, /accounts/[id]	Product account creation, automatic terms, status, balance, delivery, corrections, lifecycle-gated cancellation/reactivation documents, and payment entry.
Payments	/payments, /payments/new	Payment recording, receipt numbers, customer receipt communication, grouped history, deletion/recalculation, and staff-safe filtering.
Products	/products, /products/new, /products/[id], /products/[id]/edit	Catalog management, product economics, quantity on sale, category badges, and procurement tab.
Procurement	/products?tab=procurement	Readiness list for products with accounts at least 70% paid and still pending delivery.
Staff	/staff, /staff/new, /staff/[id], /staff/[id]/edit	Staff records, permissions, assigned work, salary tracking, deactivation, deletion, and password resets.
Staff Applications	/staff/applications	Admin review queue for signup requests.
Credits & Refunds	/credits	Open credits from overpayments or closures and refund tracking.
Reports	/reports, /api/reports/weekly-export	Admin reports, salary tracking, and Excel workbook export including procurement list.
Audit Logs	/audit-logs	Read-only trail of important system actions.
Settings	/settings, /settings/legal	Admin control panel for company, layaway, payroll, security, legal templates, addressed terms, customer communications, theme, and system metadata.
Notifications	/api/notifications	Attention counts for modules such as procurement; sidebar clears opened notifications locally.
AI Support	/api/support/assistant, components/ai-support-chat.tsx	Admin-only optional OpenAI support assistant for GLV workflows.
Health/Logout	/api/system/health, /api/logout	Basic health check and logout/cookie cleanup.

6. Customer, Account, Payment, and Delivery Flow

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&gt; procurement list quantity decreases or clears</w:t></w:r></w:p>

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Flow step	Server code	Key controls
Create customer	actions/customers.ts	Duplicate warning, staff assignment, permission/ownership checks, audit log.
Create account	actions/accounts.ts, lib/customer-account-creation.ts	Uses selected product economics, creates expected end date and opening balance.
Automatic terms	lib/customer-documents.ts, lib/customer-communications.ts	Creates addressed terms after account creation and queues one customer channel when contact details exist.
Record payment	actions/payments.ts, lib/payment-recording.ts	Rejects closed/suspended/cancelled/completed accounts, generates receipt, updates balance/status, creates overpayment credit, and

		queues one receipt channel.
Delete payment	actions/payments.ts	Recalculates account totals and balance; restricted to permitted users.
Correct account price/product	actions/accounts.ts	Admin password confirmation, recalculates balances, can create credit when product/price drops below paid amount.
Confirm delivery	actions/accounts.ts	Only after account is completed and balance is zero; revalidates account/customer/product screens.

7. Product and Procurement Module

The procurement list is a computed view, not a separate table. It groups pending-delivery customer accounts by product when their payment progress reaches the effective procurement threshold. The current rule is at least 70% paid, including fully paid accounts that are not yet delivered. Once delivery is confirmed, the account stops contributing to procurement quantity.

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5: Procurement Readiness Flow</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t
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transportCost)</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve">
+--&gt; averageProgress and highestProgress displayed</w:t></w:r></w:p><w:p><w:pPr><w:pStyle
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remove or reduce product from procurement list</w:t></w:r></w:p>
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Procurement field	Meaning
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Products to buy	Number of product groups with at least one eligible pending-delivery account.
Total units	Total eligible pending-delivery accounts across all products.
Estimated total cost	Sum of landed product cost for all units to buy.
Average paid	Average payment progress across eligible accounts for that product.
Highest paid	Highest payment progress across eligible accounts, used for sorting.
Clearing rule	Delivery confirmation changes deliveryStatus to DELIVERED; the account is excluded from procurement.

8. Staff, Applications, Salary, and Password Management

Staff can enter through an application workflow or be created directly by an admin. Approved staff can receive a user account, staff code, permissions, and a temporary/reset password. Salary tracking records monthly salary obligations and salary payments for reports.



Capability	Details
Staff list/detail	Shows staff code, contact info, active status, assigned customers, and operational work.
Applications	Admin queue for public signup requests with approve/reject outcomes.
Permissions	Granular permissions can expand staff access to admin sections without changing role to admin.

Salary	Monthly salary stored on Staff; salary payment records support report balances and payroll summaries.
Password reset	Admin reset produces temporary credentials and can require password change.
Deletion/deactivation	Destructive staff operations are admin-gated and protected by confirmation logic where applicable.

9. Credits, Refunds, Lifecycle, and Closure Rules

Credits are created when a payment overpays an account or when lifecycle closure creates a refund credit after deducting the service fee. Lifecycle refresh checks active-like accounts and moves them toward DORMANT, PROBATION, or CLOSED based on inactivity.

Rule	Current behavior
Dormant	Account becomes DORMANT after 21 days without payment activity.
Probation	Account becomes PROBATION after 4 months without payment activity.
Closed	Account becomes CLOSED after 6 months without payment activity.
Closure refund	When closing with paid amount and no prior closure credit, 32% service fee is deducted and remaining amount becomes an open credit.
Overpayment credit	Payment amount beyond remaining balance creates CustomerCredit linked to the payment/account.
Refund action	Credits can be marked refunded by permitted payment/admin users.

10. Reports, Weekly Export, and Activity

Reports combine customer/account/payment/staff/salary data into admin-facing summaries and an Excel workbook. The weekly export refreshes lifecycle statuses first, then builds sheets for executive summary, staff performance, accounts, weekly payment history, product profitability, procurement list, and weekly ledger.

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<w:p><w:pPr><w:pStyle w:val="DiagramTitle"/></w:pPr><w:r><w:rPr><w:b/></w:rPr><w:t xml:space="preserve">Diagram
7: Reporting and Export Flow</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t
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getWeeklyStaffPerformanceReport()</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t
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refreshAccountLifecycleStatuses()</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t
xml:space="preserve"> +--&gt; query staff, customers, accounts, payments, salaries,
products</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve"> +--
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Profitability</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve"> +--
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xml:space="preserve"> +--&gt; Weekly Ledger</w:t></w:r></w:p>

```

Report component	Purpose
Admin report summary	High-level business metrics, balances, status counts, salaries, collection data.
Staff performance report	Ranks staff by collection and assigned account/customer activity.
Activity report	Dashboard/activity charts and recent payment/customer/account signals.
Weekly Excel export	Downloadable workbook intended for operations review and management reporting.
Procurement sheet	Products ready to buy with quantity, costs, layaway price, average paid %, and highest paid %.

11. Notifications, Customer Communications, and AI Support

The system has computed in-app attention items for operators and persisted outbound customer messages. Customer documents and receipts queue exactly one enabled channel. Legal/document messages prefer email, then WhatsApp, then SMS; payment receipts prefer WhatsApp, then SMS, then email. With no email or phone, no outbound row is queued and staff use the in-system receipt/tracking record. Delivery preserves the selected channel and retries provider failures.

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8: Notification Clear-on-Open Flow</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t
xml:space="preserve">AppShell loads protected page</w:t></w:r></w:p><w:p><w:pPr><w:pStyle
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xml:space="preserve"> +--&gt; if user can manage products, compute procurement
list</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve">
|</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve">
v</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve">Sidebar
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xml:space="preserve"> +--&gt; click opens /products?tab=procurement</w:t></w:r></w:p><w:p><w:pPr><w:pStyle
w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve"> +--&gt; store local dismissal
signature</w:t></w:r></w:p><w:p><w:pPr><w:pStyle w:val="CodeBlock"/></w:pPr><w:r><w:t xml:space="preserve"> +--
&gt; hide same badge until count/label changes</w:t></w:r></w:p>

```

Feature	Implementation
Notification source	app/api/notifications/route.ts
Badge rendering	components/dashboard-nav.tsx
Dismissal storage	components/app-shell.tsx localStorage key glv-dismissed-attention
AI support	Admin-only floating chat bubble; route calls OpenAI only when OPENAI_API_KEY is configured.
AI support scope	Navigation, permissions, payments, accounts, products, procurement, staff, reports, settings, and troubleshooting.
Customer message queue	CustomerMessage records are created by lib/customer-communications.ts and dispatched by the notifications route or daily cron.
No-contact handling	No message is queued; staff explain terms verbally and show the receipt/product tracking information.

12. Settings Module

Settings is the broad admin control panel. It persists company profile, layaway defaults, fees, procurement threshold, payroll defaults, receipt/customer/staff code prefixes, security toggles, notification preferences, theme settings, and system status labels. Some settings are actively consumed; others are stored for future or partial wiring, so operators should distinguish saved configuration from behavior that is actually enforced downstream.

Settings group	Examples	Current use
Company profile	companyName, tradingName, tagline, phone, address, tax IDs	Used for branding and records; visible in shell and settings.

Layaway rules	installmentDurationDays, defaultDailyCollection, minimumDeposit	Used as product/account creation defaults where wired.
Procurement	procurementThresholdPercent	Default and effective procurement readiness floor is 70%.
Payroll	defaultMonthlySalary, commission fields, payrollDay	Salary defaults and reporting support.
IDs and receipts	receiptPrefix, customerIdPrefix, staffCodeLength	Receipt generation and ID/code helper logic.
Security	passwordLength, sessionTimeoutMinutes, requirePasswordChange, twoFactorEnabled	Some fields are stored; password-change enforcement is live through Auth.js/session logic.
Legal documents	editable terms/cancellation/reactivation/receipt templates	Terms are generated for new accounts and managed in Settings; lifecycle calculations are account-stage gated.
Notifications	email/sms/WhatsApp toggles	Controls the live one-channel customer queue. Provider credentials are required for delivery.
Appearance/status	theme, colors, loadingAnimation, currentVersion, databaseStatus	Stored and partly consumed in UI branding/status surfaces.

13. Auditability and Data Safety

The system records audit logs for major business writes. Important destructive or financial operations are gated by roles, permissions, staff ownership checks, admin password confirmation, confirmation text, or transaction boundaries. Staff-facing data is shaped server-side so restricted financial totals are not merely hidden in the UI.

Control	Where it appears
Server-side auth	auth config, page/session checks, server actions
Permissions	UserPermission enum and hasPermission checks
Ownership restrictions	Staff can act on assigned customers/accounts unless broader permission exists
Transactions	Payments, account corrections, staff/customer/product destructive flows, lifecycle closure
Audit logs	Create/update/delete/payment/lifecycle/settings/staff actions
Credential safety	Secrets are environment variables; documentation must never print values.

14. Deployment, Verification, and Operations

Command	Purpose	Current note
npm run lint	ESLint project check.	Passed in latest verification.
npx tsc --noEmit	TypeScript check without emitting files.	Passed in latest verification.
npx next build	Next production build without migration deploy.	Passed in latest verification.
npm run build	Runs prisma generate and next build.	Does not deploy migrations.
npm run db:deploy	Deploy migrations explicitly before a dependent release.	Use DATABASE_URL_UNPOOLED from a trusted operator environment; do not use the port 6543 transaction pooler.
npm run seed	Run Prisma seed script.	Use only when intentionally bootstrapping data.

Environment variable names used by the system include DATABASE_URL, AUTH_SECRET, AUTH_URL, OPENAI_API_KEY, and OPENAI_MODEL. Values are intentionally omitted from this document.

15. Important Caveats and Known Follow-Ups

Item	Current status / recommendation
Screenshots	Live authenticated screenshots were not embedded because the in-app browser connector failed in this Windows session before opening. Diagrams are included instead.
Settings effectiveness	Many settings persist today; verify downstream consumers before claiming a setting changes behavior.
Notification persistence	Notifications clear locally by signature; there is no database notification/read model yet.
AI support persistence	Chat state is browser-local and not audited or stored.
Prisma package config	Prisma warns that package.json#prisma is deprecated for Prisma 7; not currently blocking.
Database reachability	Full build and migration deploy require Supabase connectivity through the correct pooler.
Unstaged local file	app/page.tsx had an unrelated local edit at the time this documentation was generated; it is not part of the pushed procurement/reporting release.

16. Automatic SMS Notifications

BMS Africa provides SMS using the mNotify Quick Bulk SMS API. Configure MNOTIFY_API_KEY, MNOTIFY_SENDER_ID (approved GLV sender GODS LOVE V), and CRON_SECRET server-side. Apply the

SmsNotification migration before enabling SMS Notifications in Settings. Never expose the API key or log request URLs containing it.

Super administrators configure five database-backed message templates on Settings > SMS: Salary payment, Customer welcome, 70% progress, Missed payment, and Weekly payment summary. The page lists the placeholders permitted for each event, renders a sample preview, validates a 612-character maximum, and can restore all defaults. Saved templates apply to newly queued messages; existing queue rows keep their original message snapshot. Salary resolves to the staff member on the salary payment. Customer templates resolve to the qualifying customer. Invalid or missing phone numbers fail visibly and are never sent to a fallback recipient.

Default message text is branded Rock Frost Group, while every staff/customer name placeholder renders only the first whitespace-delimited name. The BMS handset sender is Rock Frost, which fits the provider's 11-character limit and must be approved in the BMS account before production configuration.

A saved salary payment queues one staff SMS. A new product account queues one welcome SMS at its start date. Payment totals at or above 70 percent queue one milestone SMS per account. Payment edits also check the milestone. No historical salary or welcome messages are backfilled. These alerts always use SMS, independently of the existing document and receipt channel preference.

Recording a staff deposit triggers the weekly customer summary. For that staff member, GLV totals each customer's Payment records from Monday through Sunday across all product accounts and queues one summary only for customers whose total is greater than zero. The dedupe key combines staff, customer and week. Another deposit cannot duplicate an accepted message, but it can refresh a summary that is still Pending or Failed. Customers never receive another staff member's totals.

Active and overdue accounts with a positive balance qualify for reminders after seven full days without payment, and at most once per unpaid week thereafter. The later of account start, latest payment date and latest payment entry time anchors the interval. Backdated entries restart it. Other lifecycle statuses are excluded. The Missed payment template supports staffName, resolved to the first name of the staff member assigned to that customer. The protected /api/cron/sms-notifications job runs at 09:00 UTC daily, queues reminders and dispatches up to 100 messages in groups of five. Authenticated activity and qualifying mutations also dispatch pending messages. Monitor backlog and schedule more frequent runs if volume requires it.

The dedicated /settings/sms page lets super administrators enable or pause automatic SMS, verify provider/API/sender configuration, review the five notification rules, inspect the latest 100 queue records, and retry definitive failures. The queue records unique event keys within the business transaction. Conditional claims prevent concurrent workers sending the same item. Before dispatch, source existence, account eligibility and current phone are checked. Correct missing phones or provider errors before Retry. Provider acceptance is labelled ACCEPTED, not confirmed handset delivery. Check BMS campaign history using the saved campaign ID. Rate limits retry after one hour, up to five attempts.

Timeouts and interrupted workers become UNKNOWN and require reconciliation rather than blind resending. Restored pending or processing messages become UNKNOWN. Disabling SMS pauses the queue.

Implementation checks use mocked provider/database tests in scripts/sms.test.ts, TypeScript, lint and a build without migration deployment. Production migration, credentials, authenticated UI and real delivery must be verified separately before activation is considered complete.

Appendix A. Source File Map

Area	Representative files
Pages	app/dashboard/page.tsx; app/customers/page.tsx; app/accounts/page.tsx; app/payments/page.tsx; app/products/page.tsx; app/staff/page.tsx; app/reports/page.tsx; app/settings/page.tsx
Server actions	actions/customers.ts; actions/accounts.ts; actions/payments.ts; actions/products.ts; actions/staff.ts; actions/settings.ts; actions/salaries.ts; actions/applications.ts
Domain libraries	lib/auth.config.ts; lib/reports.ts; lib/weekly-excel-report.ts; lib/procurement.ts; lib/payment- recording.ts; lib/customer-account-creation.ts; lib/account-lifecycle.ts
UI components	components/app-shell.tsx; components/dashboard-nav.tsx; components/customer-form.tsx; components/account-form.tsx; components/payment-form.tsx; components/product-form.tsx
API routes	app/api/notifications/route.ts; app/api/reports/weekly-export/route.ts; app/api/support/assistant/route.ts; app/api/change-password/route.ts; app/api/system/health/route.ts
Data model	prisma/schema.prisma; prisma/migrations/*